**ASSESSMENT-2**

**CSA1707- ARTIFICAL INTELLIGENCE**

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**Question 1: AI-Powered Drone for Emergency Medicine Delivery**

**Problem Understanding**

A government deploys AI-powered drones to deliver emergency medicines in flood-affected regions. The environment is highly dynamic because roads may become blocked unexpectedly, weather conditions change travel costs, and the drone receives only partial real-time updates. The drone must choose routes that minimize delivery time, travel cost, and risk while ensuring medicines reach affected people safely.

**Method Selected**

The problem involves **Greedy Best-First Search** and **A* Search**, which are informed search algorithms. Both use heuristic information, but they differ in how they evaluate paths.

**(i) Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty.**

**Answer**

Greedy Best-First Search selects the path that appears closest to the destination based only on the heuristic value (straight-line distance). It ignores the actual travel cost and focuses on reaching the goal as quickly as possible.

**Strengths**

- Makes decisions quickly.
- Requires less computation.
- Suitable when immediate response is required.

**Weaknesses**

- Ignores actual travel cost.
- May choose blocked or risky routes.
- Does not always produce the optimal path.
- Performs poorly when the environment changes frequently.

**(ii) Explain how A* would handle the same situation, including how it balances cost and heuristic.**

**Answer**

A* Search evaluates each path using:

**f(n) = g(n) + h(n)**

where:

- **g(n)** = Actual cost travelled.
- **h(n)** = Estimated distance to the destination.

By considering both actual cost and heuristic information, A* avoids expensive or blocked routes and usually finds the safest and shortest path.

**(iii) Analyze the impact of heuristic accuracy on both algorithms.**

**Answer**

The quality of the heuristic greatly affects the performance of both algorithms.

- If the heuristic is accurate, both algorithms reach the destination more efficiently.
- If the heuristic is inaccurate, Greedy Best-First Search may choose poor routes.
- A* is less affected because it also considers the actual path cost.
- An admissible heuristic allows A* to guarantee the optimal solution.

**(iv) Discuss how dynamic changes (blocked paths) affect search performance.**

**Answer**

Blocked paths require the drone to update its route during navigation.

- Greedy Best-First Search may repeatedly choose blocked paths because it relies mainly on the heuristic.
- A* can recompute a better route by considering updated path costs and heuristic values.
- Frequent environmental changes increase computation but improve safety and reliability.

**(v) Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety.**

**Answer**

**Recommended Algorithm: A* Search**

**Justification**

A* Search is the most suitable algorithm because it balances the actual travel cost and heuristic estimate. It avoids blocked or costly routes, adapts better to dynamic environments, and provides an optimal path when using an admissible heuristic. Although it requires more computation than Greedy Best-First Search, it offers greater safety, reliability, and efficiency, which are critical for emergency medicine delivery.

**Conclusion**

For emergency drone delivery in a dynamic flood-affected environment, **A* Search** is the preferred algorithm. It provides better decision-making by balancing speed, optimality, and safety while adapting to changing environmental conditions.

**Question 2: AI-Based Traffic Signal Optimization**

**Problem Understanding**

A smart city uses an AI system to optimize traffic signals across hundreds of intersections. The objective is to minimize **waiting time, fuel consumption, and traffic congestion** while continuously adapting to changing traffic conditions. Since the search space is extremely large and traffic patterns change frequently, traditional search methods become inefficient.

**Method Selected**

This problem is an **Optimization Problem**. The suitable AI techniques are **Hill Climbing**, **Simulated Annealing**, and **Genetic Algorithms**.

**(i) Formulate this as an optimization problem and explain why traditional search is inefficient.**

**Answer**

This is an optimization problem where the objective is to find the best traffic signal timings that minimize waiting time, fuel consumption, and traffic congestion.

Traditional search algorithms are inefficient because:

- The number of possible signal combinations is extremely large.
- Traffic conditions change continuously.
- Searching every possible solution requires excessive time and computation.
- Real-time decision making becomes difficult.

Therefore, optimization algorithms are preferred over traditional search methods.

**(ii) Explain how Hill Climbing would work in this scenario and identify its limitations.**

**Answer**

Hill Climbing starts with an initial traffic signal schedule and continuously modifies it to improve traffic flow. At each step, it selects the neighboring solution that provides the greatest improvement.

**Advantages**

- Simple to implement.
- Produces solutions quickly.
- Suitable for real-time optimization.

**Limitations**

- May get stuck in local maxima.
- Cannot guarantee the global optimum.
- Performs poorly when many similar solutions exist.

**(iii) Discuss issues like local maxima, plateaus, and ridges with real-world interpretation.**

**Answer**

**Local Maxima:**
The algorithm finds a good traffic signal plan but misses a better one because no neighboring solution appears better.

**Plateaus:**
Several signal timings produce nearly identical traffic flow, making it difficult for the algorithm to decide which direction to move.

**Ridges:**
The optimal solution requires multiple coordinated changes across intersections, but Hill Climbing changes only one solution at a time, making progress slow.

**(iv) Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations.**

**Answer**

**Simulated Annealing** occasionally accepts worse solutions during the search, allowing it to escape local maxima and explore better regions of the search space.

**Genetic Algorithms** generate multiple candidate solutions and improve them using selection, crossover, and mutation. This increases the chance of finding the global optimum and handles large search spaces effectively.

**(v) Recommend the best approach and justify your choice based on scalability and adaptability.**

**Answer**

**Recommended Approach: Genetic Algorithm**

**Justification**

A Genetic Algorithm is the most suitable approach because it efficiently searches large solution spaces, adapts to changing traffic conditions, and avoids getting trapped in local maxima. It can optimize traffic signals across hundreds of intersections simultaneously while producing high-quality solutions within a reasonable time.

**Conclusion**

Traffic signal optimization is a large-scale optimization problem. Among the available techniques, the **Genetic Algorithm** provides the best balance of scalability, adaptability, and solution quality, making it highly suitable for smart city traffic management systems.

**Question 3: Autonomous Mars Rover – Online Search Agent**

**Problem Understanding**

An autonomous rover is deployed on Mars to explore unknown terrain. It must safely navigate, collect samples, avoid hazards such as rocks and craters, and operate with limited sensing capability. Since the rover does not have a complete map, it must update its knowledge and make decisions while exploring in real time.

**Method Selected**

The most suitable approach is an **Online Search Agent**, as the rover explores an unknown and dynamic environment where complete information is unavailable.

**(i) Explain why this problem requires an online search agent instead of offline search.**

**Answer**

An Online Search Agent is suitable because the rover does not know the complete environment before starting its mission. It gathers information while moving and updates its decisions based on newly discovered terrain. In contrast, an Offline Search Agent requires a complete map before planning a path, which is not possible on Mars.

**(ii) Analyze the challenges of partial observability and dynamic environments.**

**Answer**

The rover faces several challenges:

- It cannot observe the entire terrain at once.
- New obstacles such as rocks and craters are discovered only during exploration.
- Communication delays prevent immediate assistance from Earth.
- The rover must continuously update its decisions while ensuring safe navigation.

These factors make the environment partially observable and dynamic.

**(iii) Describe how the rover builds and updates its internal model.**

**Answer**

The rover uses data collected from its sensors to create an internal map of the environment. As it moves, it records safe paths, identifies obstacles, and updates its map with newly discovered information. This internal model helps the rover avoid revisiting dangerous areas and improves future navigation decisions.

**(iv) Compare online search with uninformed and informed search strategies.**

**Answer**

| **Online Search** | **Uninformed Search** | **Informed Search** |
|---|---|---|
| Learns while exploring | Uses no heuristic information | Uses heuristic information |
| Suitable for unknown environments | Requires predefined search space | Requires heuristic estimates |
| Updates decisions continuously | Explores systematically | Searches efficiently using heuristics |
| Best for real-time exploration | Better for known environments | Best when sufficient information is available |

**(v) Justify the most suitable approach considering uncertainty, adaptability, and safety.**

**Answer**

**Recommended Approach: Online Search Agent**

**Justification**

An Online Search Agent is the best choice because it adapts to unknown environments, updates its knowledge continuously, and makes decisions using real-time sensor information. It safely avoids hazards while navigating and can respond effectively to unexpected obstacles. These features make it highly suitable for autonomous Mars exploration.

**Conclusion**

An **Online Search Agent** is the most appropriate approach for the Mars rover because it supports real-time decision-making, adapts to unknown environments, and ensures safe and efficient exploration despite limited information.